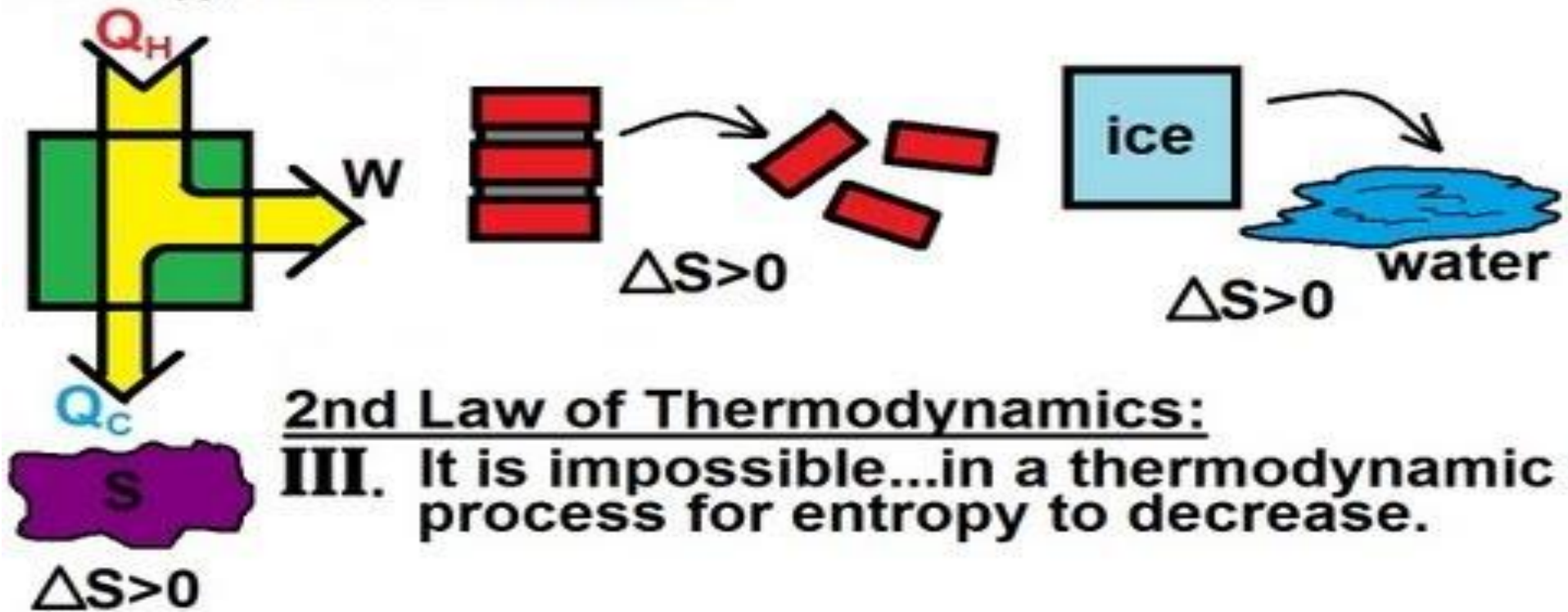


S=Entropy: A measure of the amount of energy this is unavailable for work
S=Entropy: A measure of disorder



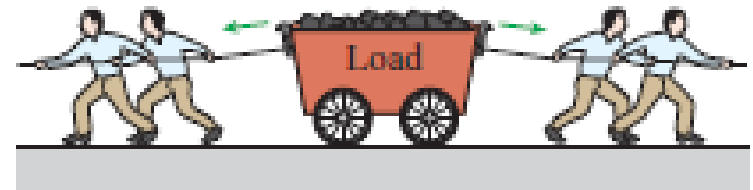
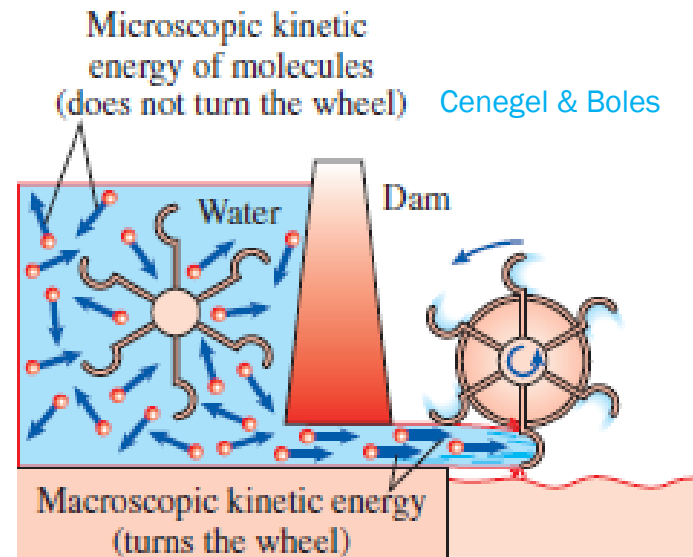
Module-5: Entropy

Thermodynamics (NMEC103) : (Winter 2024-25)

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Entropy: As 1st law applied to a process gives birth to a thermodynamic property called energy (E), application of 2nd law to a process introduces another extensive property called entropy (S). Although entropy may be defined as a measure of molecular disorderness or randomness or chaos, this definition is not of much use. Hence In order to understand entropy the focus should be on “What is it used for ?” and “How is it used?” instead “What is entropy?”

It has already been discussed that work, Bulk KE & PE etc. are a ordered form and heat and internal energy are of disordered form of energy. The following figure will put further light on the ordered ness. The microscopic kinetic energy which is a disordered form of energy is unable to turn the wheel, however, the macroscopic bulk KE is organised. Hence is more useful. *So higher the orderedness higher will be the work potential*



Disorganized energy does not create much useful effect, no matter how large it is.

When ice (highly ordered molecular structure) is heated it melts (becomes water), the intermolecular force of attraction reduces, if water (amorphous molecular structure) will further heated it will become steam (highly random). Ice was having a definite shape and volume, water is having a definite volume but no definite shape but steam has neither. So it can be concluded that when ice, becomes water and water becomes steam the randomness increases (it is more difficult to specify the position, velocity of the molecules in the gaseous phase compared to that of solid phase). This is in fact true, for at the triple point of water (the only point where the solid liquid and vapor phases can coexists in equilibrium.)

Specific entropies of these phases at triple point of water are:

$$(\text{Specific entropy})_{\text{solid}} = -1.221 \text{ kJ/kg.K}$$

$$(\text{Specific entropy})_{\text{liquid}} = 0.0 \text{ kJ/kg.K}$$

$$(\text{Specific entropy})_{\text{vapor}} = 9.157 \text{ kJ/kg.K}$$

Even though the physical concept of entropy presented here is based on the molecular (or microscopic) behavior of the system, order/disorder phenomena exists at all levels macroscopic as well as microscopic. For example, a very messy, disordered room can be said to have a high entropy value.

It can be shown that entropy also increases in the same direction i.e. *Entropy follows the direction of increase in randomness*

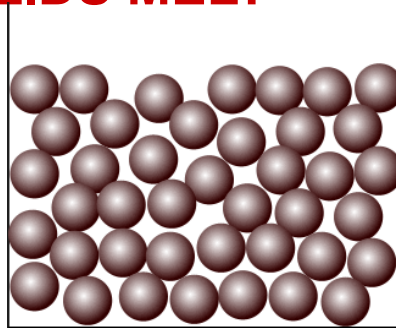
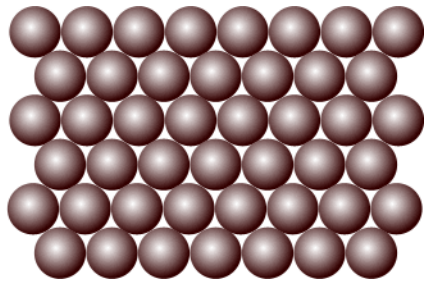
Entropy, disorder, probability, spontaneous process & Time

Entropy increases when

- solids melt
- liquids boil
- ionic solids dissolve in water
- the number of gas molecules increases
- the temperature increases

Entropy increases when

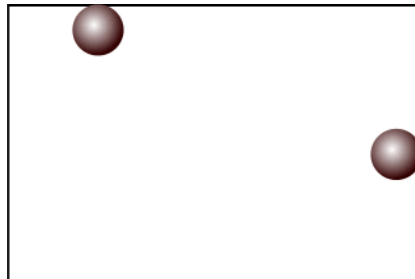
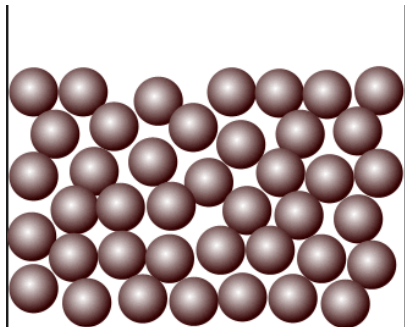
SOLIDS MELT



Melting of Ice

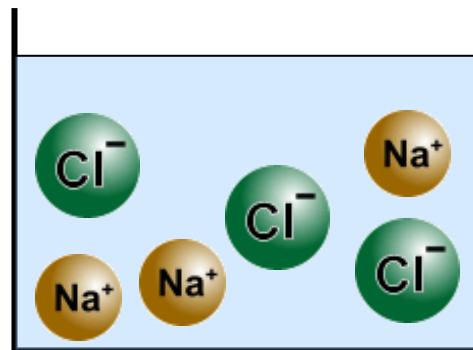
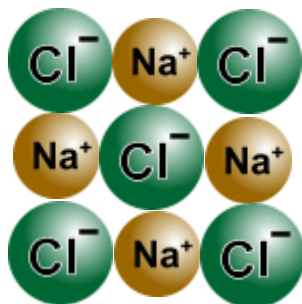
Entropy increases when

LIQUIDS BOIL



Entropy increases when

IONIC SOLIDS DISSOLVE IN WATER

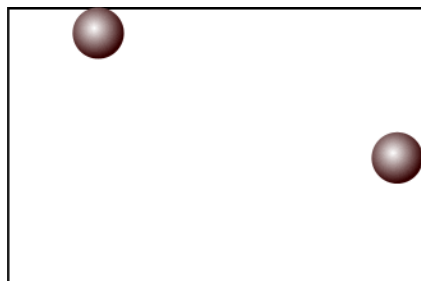


Regular arrangement in
an ionic crystal lattice

Ions dissociate in water;
there is less order

Entropy increases when

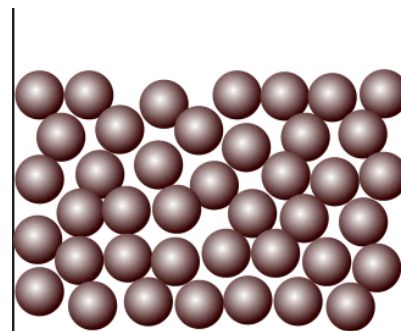
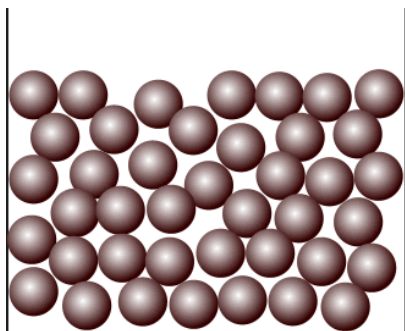
THE MOLES OF GAS INCREASE



Particles in gases move in a
random way. The more gas
molecules there are, the greater
the degree of randomness.

Entropy increases when

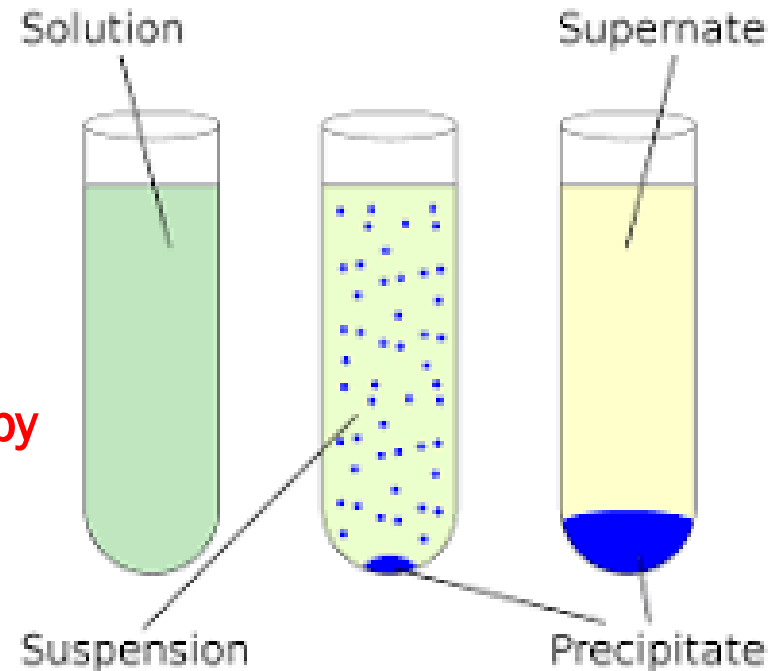
THE TEMPERATURE INCREASES



In chemical reaction if the **no of moles in the product side is higher than that of the reactant side**, then **entropy increases and vice-versa**. When oxygen converts to ozone

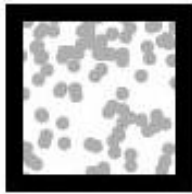


- When **precipitation occurs**, entropy decreases.
- When **rubber is stretched** its molecules become more organized and entropy decreases
- **A cold crystalline solid has the minimum entropy**
- **If graphite will be converted to diamond**, then entropy decreases
- ❖ The entropy of a system can not be infinite as a finite system can not have infinite microstates
- When **egg boils** the protein denaturizes in which molecules spread down and the water present also evaporates so entropy increases

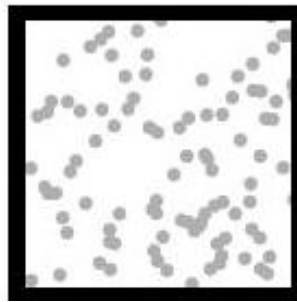


destroy the characteristic properties of (a protein or other biological macromolecule) by heat, acidity, or other effect which disrupts its molecular conformation.

(a)

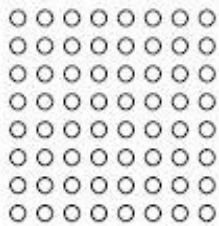


expand
→

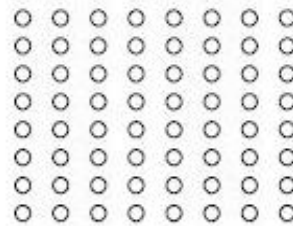


Isothermal Expansion of Gas
entropy **increases**

(b)



stretch
→



Isothermal Stretching of a
Metallic Rod
entropy **increases**

(c)



stretch
→

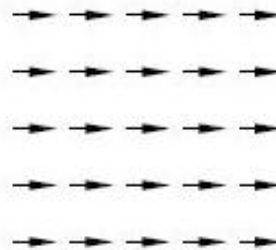


Isothermal Stretching of Rubber
entropy *decreases*

(d)



magnetize
→

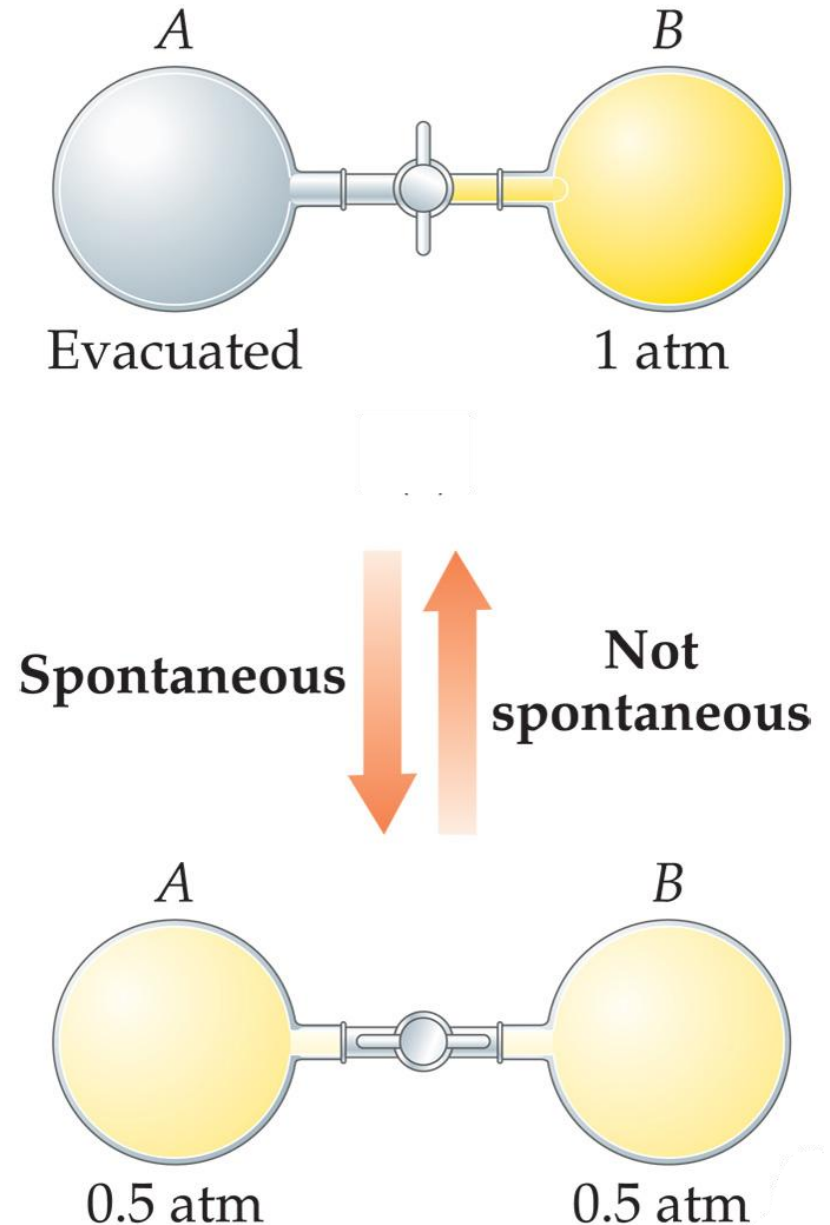


Isothermal Magnetization of a
Paramagnetic Substance
entropy *decreases*

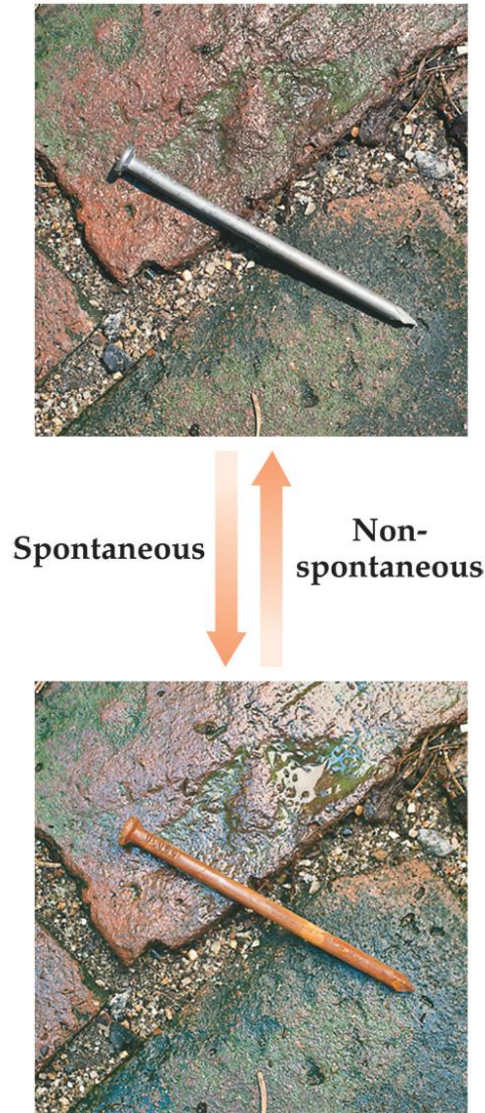
Spontaneous Processes: Do not requires **external effort**

May be slow or Fast Process

- Spontaneous processes are those that can proceed without any outside intervention.
- The gas in vessel *B* will spontaneously effuse into vessel *A*, but once the gas is in both vessels, it will *not* spontaneously



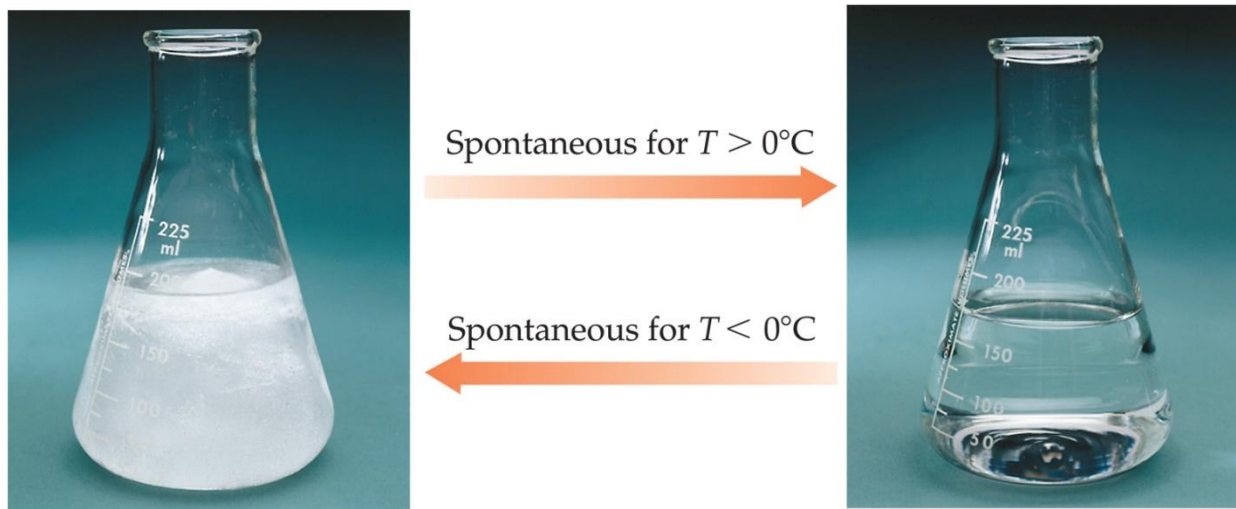
Spontaneous Processes



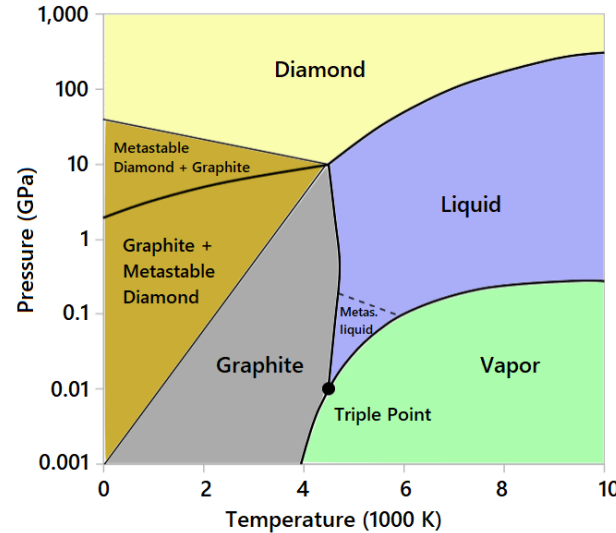
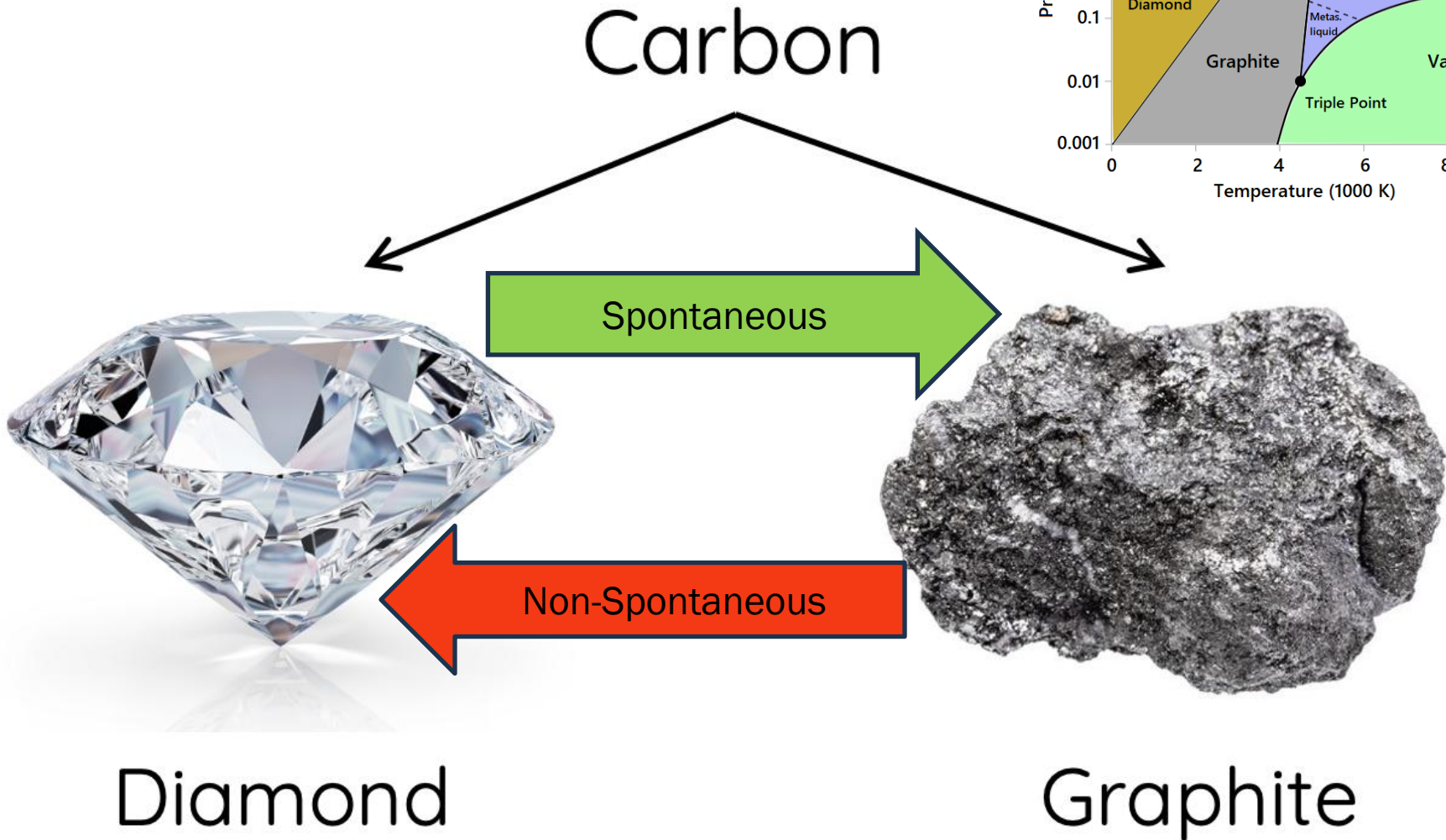
Processes that are spontaneous in one direction are nonspontaneous in the reverse direction.

Spontaneous Processes

- Processes that are spontaneous at one temperature may be non-spontaneous at other temperatures.
- Above 0°C it is spontaneous for ice to melt.
- Below 0°C the reverse process is spontaneous.



Spontaneous Processes



Probability of Disorder: Is an ordered state more probable or a disordered state?
Is there a higher probability that your room will be messy or neat as time goes on?



Which of the following two cases is more probable and which one is more disordered



The first case is **definitely more probable** and also be of **more disorderliness**. Hence , the message of the present slide is that **a disordered state or a high entropy state is a more probable state**. Hence there exists a relationship between entropy and thermodynamic probability.

Entropy and Quality

As already discussed in the context of 2nd law that energy has both quantity and quality. The internal energy U is a measure of the quantity of energy that a system possesses, S is a measure of the quality of that energy: low entropy means high quality; high entropy means low quality. As already mentioned mechanical work, kinetic energy, potential energy etc. are high grade energy, and they are because of an organised motion of the particles, do not carry any entropy while crossing the system boundary and are completely available as useful work however heat which is a disordered form of energy always carries entropy with it while entering or leaving a system and is not completely available as useful work.

Entropy and Time

The concept of entropy has importance in the discussion of time, a topic logically avoided in thermodynamics. The increase in entropy (disorder) of a system and its surroundings occur in all aging process, whether in a living organism or in an inanimate system. There is no doubt which condition is prior in time: youth or maturity, green apple or over ripen apple, structure or erosion. One condition always precedes other. Order precedes disorder. Entropy always increases during an aging process. So, in a metaphorical sense entropy is the arrow of time. Furthermore, the arrow has only one direction: it points forward into the future

Statistical Interpretation of Entropy

When viewed microscopically from a statistical thermodynamics point of view, an isolated system that appears to be at a state of equilibrium actually exhibits a high level of activity because of the continual motion of the molecules. To each state of macroscopic equilibrium, there corresponds a large number of molecular microscopic states or molecular configurations. Boltzmann first hypothesized that the entropy of a system at a specified macro state is related to the total number of possible relevant microstates of that system, w (from *Wahrscheinlichkeit*, the German word for 'probability'). This thought was formulated later by Plank using a constant k with the entropy unit of J/K named after Boltzmann (and inscribed on Boltzmann's tombstone) as

$$S = k \ln w$$

This is known as the **Boltzmann relation**. The thermal motion randomness or disorder, as related to entropy is later generalized by Gibbs as a measure of the sum of all microstates' uncertainties.

Distribution for $N = 7$ and $E = 3$

Distribution	n_0	n_1	n_2	n_3	W	W/W_{tot}
Macrostate	Microstates					
(1)	6	0	0	1	7	0.08
(2)	5	1	1	0	42	0.50
(3)	4	3	0	0	35	0.42
Total		Most Probable State			84	1.00

Distribution for $N = 50$ and $E = 5$

Distribution Macrostate	n_0	n_1	n_2	n_3	n_4	n_5	W	W/W_{tot}
(1)	49	0	0	0	0	1	50	0
(2)	48	1	0	0	1	0	2450	0
(3)	48	0	1	1	0	0	2450	0
(4)	47	2	0	1	0	0	58,800	0.02
(5)	47	1	2	0	0	0	58,800	0.02
(6)	46	3	1	0	0	0	9,21,200	0.29
(7)	45	5	0	0	0	0	21,18,760	0.67
Total							31,62,510	1.00

As N increases $W_{max} \rightarrow W_{total}$ $S = k_B \ln W_{total} \cong k_B \ln W_{max}$

Different distributions will be referred to as macrostates or configurations; different ways of realizing the same distribution or macrostate will be referred to as microstates or complexions. W now becomes the number of complexions for a given configuration. The most probable macrostate is the one with the greatest number of microstates. Macrostates are distinguishable experimentally ; microstates are indistinguishable. All microstates are equiprobable. It is only the large number of microstates associated with the most probable macrostate which makes the latter so much more probable than any other macrostate. When the number of element increases the number of microstate for the most probable case approaches total number of microstates $W_{total} \cong W_{max}$

Suppose two system which are isolated from each other have entropies S_1 & S_2 and have microstates W_1 & W_2 respectively then the combined system has entropy $S_1 + S_2$ and microstates $W_1 \times W_2$ respectively. This condition is satisfied by the expression $S = k_B \ln W$

- Another way: the **average assembly property Y_{AV}** is equal to the *value of the property Y_{MP} associated with the most probable macrostate*.
- Determination of **a given property for the most probable macrostate is equivalent to determination of the average assembly property**.

Postulational Approach to Thermodynamics

Postulate-I: “There exist particular states (called equilibrium states) of simple systems, that macroscopically, are characterised completely by the internal energy U , the volume V , and the mole numbers $N_1, N_2, \dots, N_r$ of the chemical components.”

Postulate-II: There exist a function (called the entropy S) of extensive parameters of any composite system, defined for all equilibrium states and having the following property. The values assumed by the extensive parameters in the absence of an internal constraint are those that maximize the entropy over the manifold of constrained equilibrium state.

Postulate-III: The entropy of a composite system is additive over the constituent sub systems. The entropy is continuous and differentiable and is a monotonically increasing function of energy

Postulate-IV: The entropy of any system vanishes in the state for which

$$\left(\frac{\partial U}{\partial S} \right)_{V,N} = 0 \quad \text{that is, at zero of temperature}$$

Note that the internal energy and entropy play a dominant role in this postulates. Contained within these postulates are the zeroth, First, second and third laws, and the state postulate found in the conventional thermodynamic textbook based on inductive logic.

Some Insight into thermodynamic Microstate and Macro state

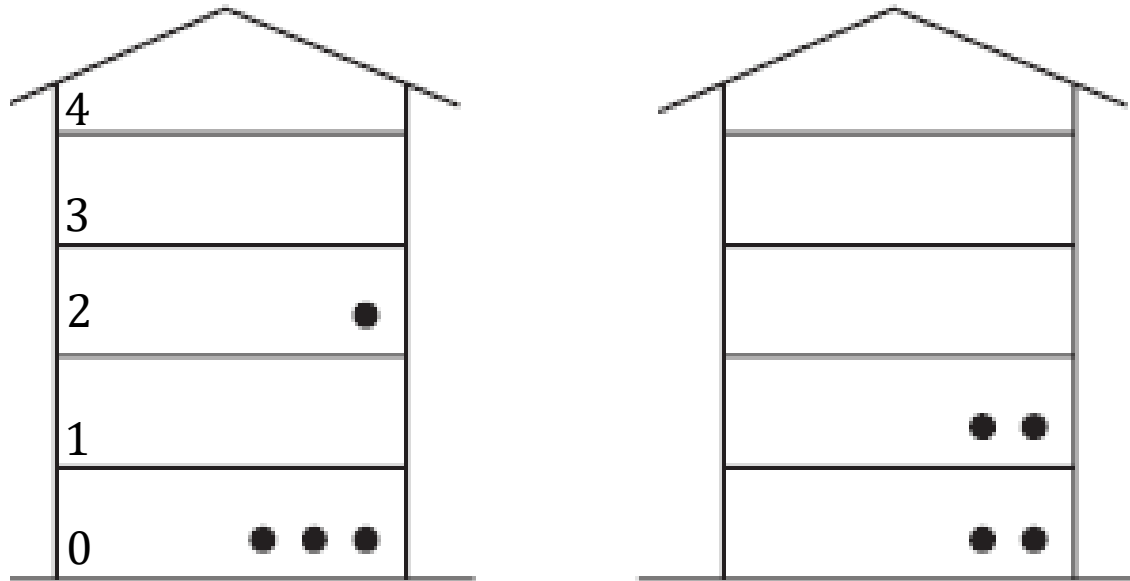
Properties like U and S for a substance at a given state are averaged over many particles on the molecular level, so they (atoms and molecules) do not all exist in the same detailed quantum state. There are a number of different configurations possible for a given state that constitutes an uncertainty or chaos in the system. The number of possible configurations, w , is called the *thermodynamic probability*, and each of these is equally possible; this is used to define the entropy as

$$S = k \ln w$$

where k is the Boltzmann constant, and it is from this definition that S is connected to the uncertainty or chaos. The larger the number of possible configurations is, the larger S is. For a given system, we would have to evaluate all the possible quantum states for kinetic energy, rotational energy, vibrational energy, and so forth to find the equilibrium distribution and w . Without going into those details, which is the subject of statistical thermodynamics, a very simple example is used to illustrate the principle'

Some Insight into thermodynamic Microstate and Macro state

Assume we have **four identical objects** that can **only possess one kind of energy**, namely, **potential energy associated with elevation (the floor) in a tall building**. Let the four objects have a combined 2 units of energy (**floor height times mass times gravitation**). How can this system be configured? We can have **one object on the second floor and the remaining three on the ground floor**, giving a total of 2 energy units. We could also have **two objects on the first floor and two on the ground floor**, again with a total of 2 energy units. **These two configurations are equally possible**, and we could therefore see the system 50% of the time in one configuration and 50% of the time in the other; we have some positive value of S .



Now let us add 2 energy units by heat transfer; that is done by giving the objects some energy that they share. Now the total energy is 4 units, and we can see the system in the following configurations (a–e):

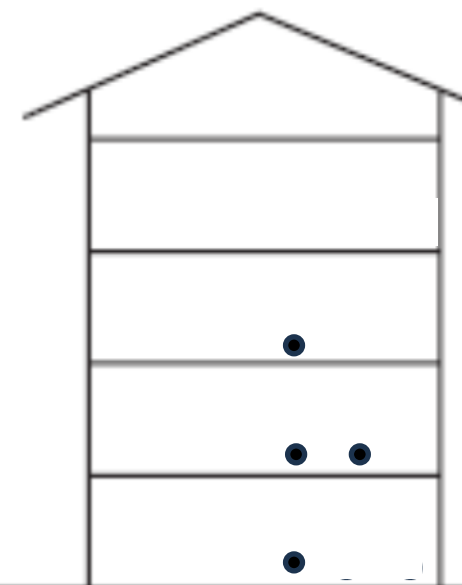
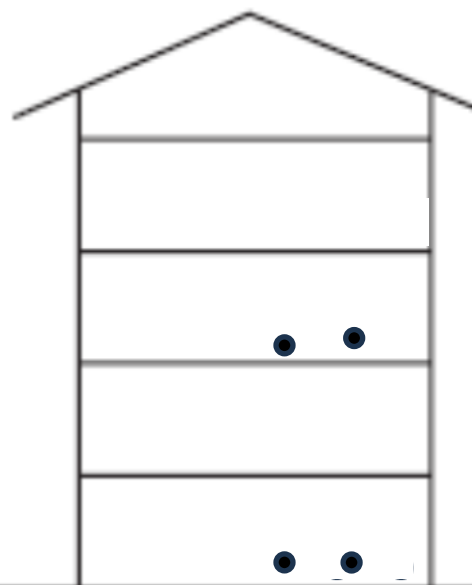
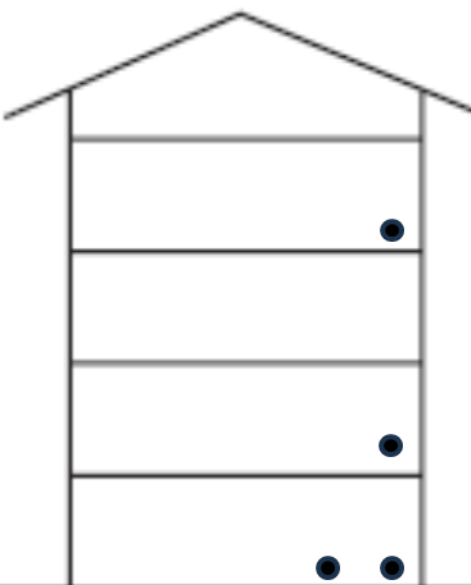
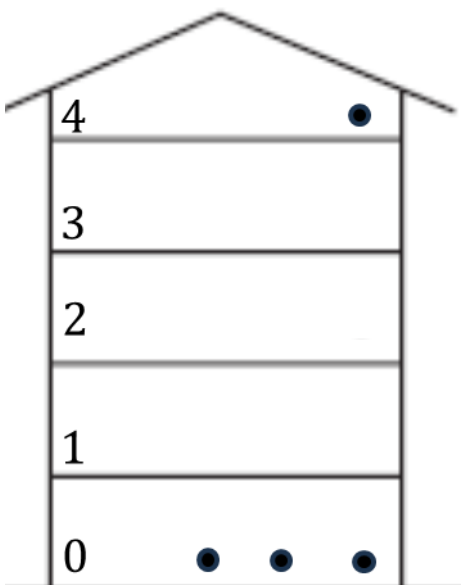
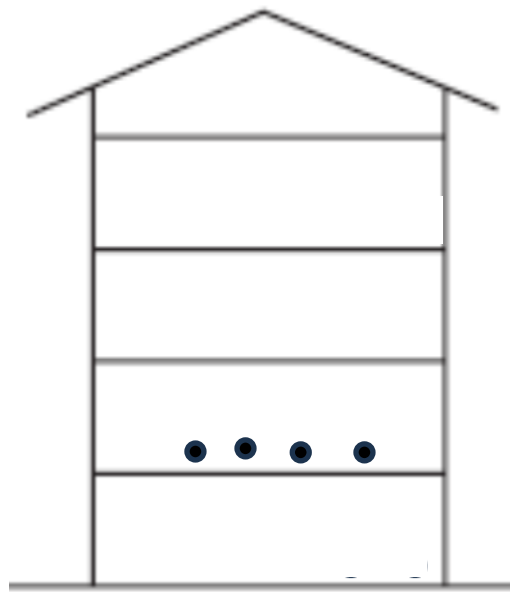
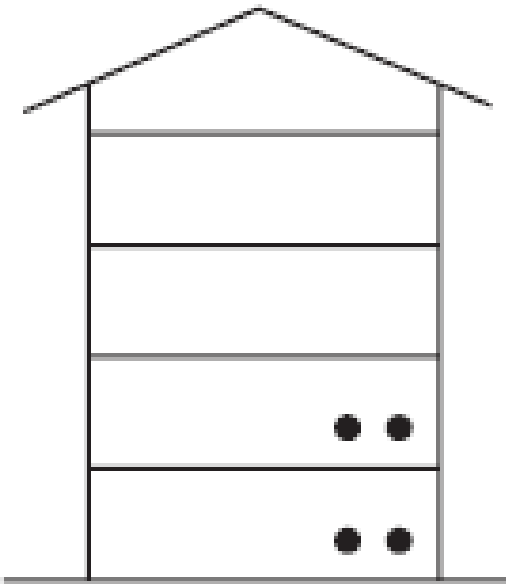
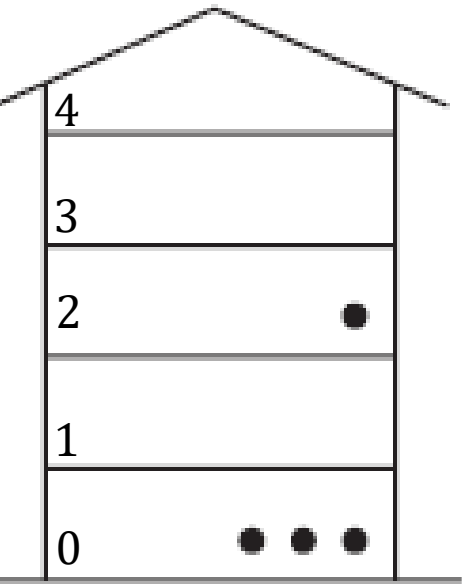
Floor number:		0	1	2	3	4
Number of objects	a:	3				1
Number of objects	b:	2	1		1	
Number of objects	c:	2		2		
Number of objects	d:	1	2	1		
Number of objects	e:		4			

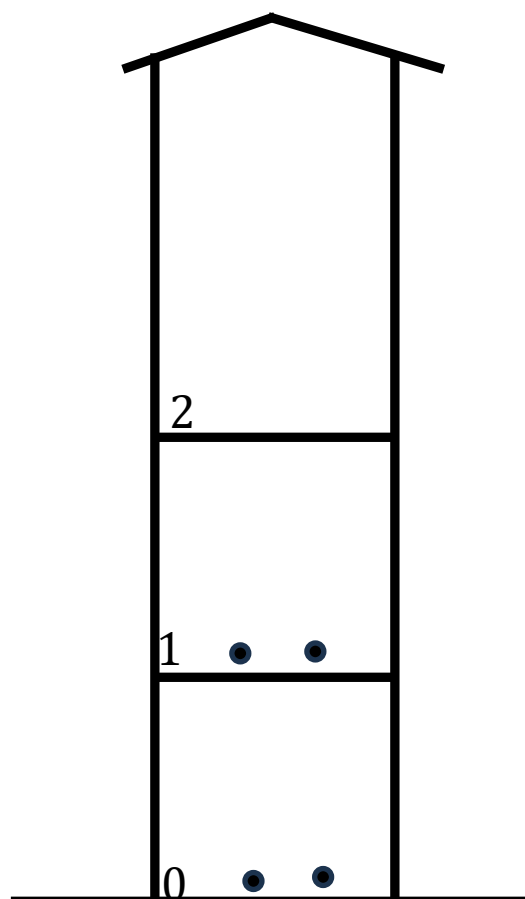
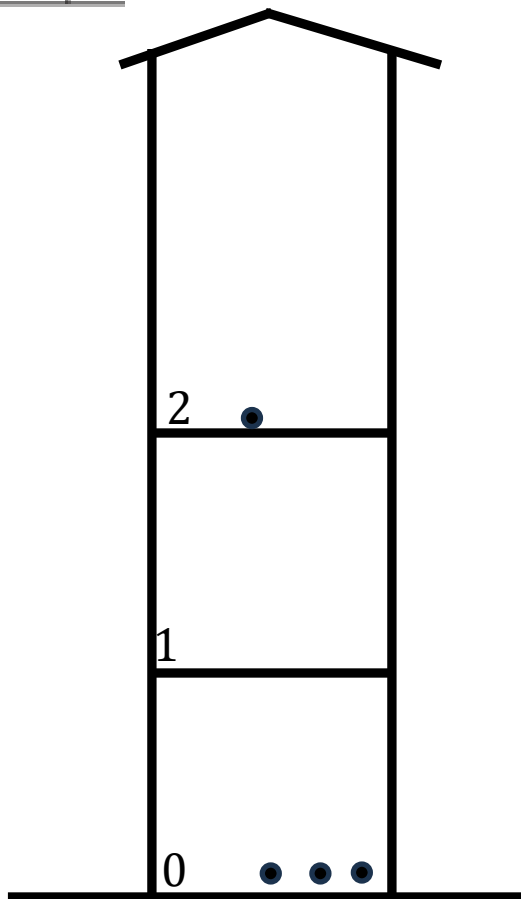
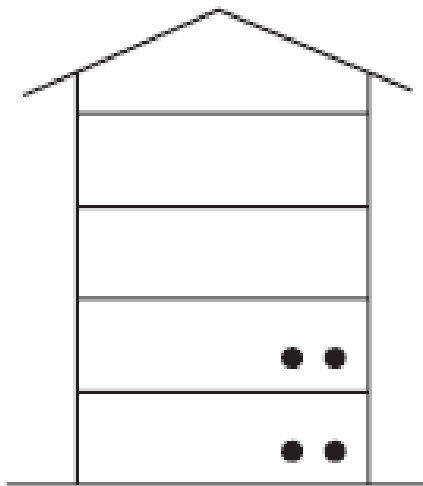
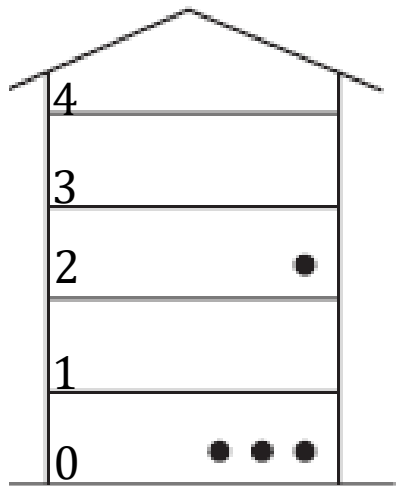
Now we have five different configurations ($w = 5$)—each equally possible—so we will observe the system 20% of the time in each one, and we now have a larger value of S .

On the other hand, if we increase the energy by 2 units through work, it acts differently. Work is associated with the motion of a boundary, so now we pull in the building to make it higher and stretch it to be twice as tall, that is, the first floor has 2 energy units per object, and so forth, as compared with the original state. This means that we simply double the energy per object in the original configuration without altering the number of configurations, which stay at $w = 2$. In effect, S has not changed.

Floor number:		0	1	2	3	4
Number of objects	f:	3		1		
Number of objects	g:	2	2			

This example illustrates the profound difference between adding energy as a heat transfer changing S versus adding energy through a work term leaving S unchanged. In the first situation, we move a number of particles from lower energy levels to higher energy levels, thus changing the distribution and increasing the chaos. In the second situation, we do not move the particles between energy states, but we change the energy level of a given state, thus preserving the order and chaos.

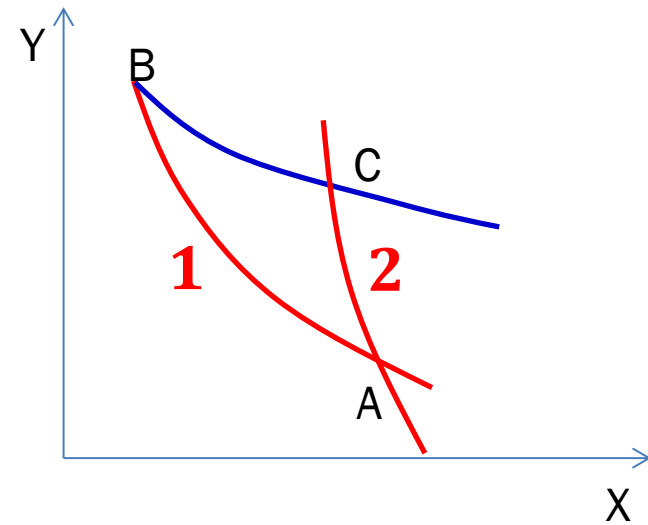




Reversible Adiabatic Process and Entropy

Applying the 2nd law of thermodynamics it can be shown that **two reversible adiabatic will never intersect**

Let's assume that the above statement is not true, then let **two reversible adiabatic intersect each other at a point say "A"**. Then an isotherm can be constructed such that it intersects these two curves at "B" and C "respectively. Now these three processes construct a thermodynamic cycle A-B-C and the area bounded by these curves is the net work/net heat interaction. **If we closely observe we can understand that this a thermodynamic cycle which interacts with single thermal reservoir during the isothermal B-C process as other two processes are adiabatic and produces net work output that is the area bounded by the curves A-B-C. this violates the K-P statement of second law**, hence our initial assumption that two reversible adiabatic will intersect is wrong. Hence two reversible adiabatic will never intersect



It is a well-known fact that two constant property line (e.g. two isobars, two isotherms or two constant volume lines) will never intersect each other as a system can not have two values of a property (say pressure) at a particular state. We will also come to know that along a reversible adiabatic line also one property called entropy remains constant. As a system can not have two entropy values at a particular state, the same has been ensured by the above proof.

V.I.M.P. Reversible adiabatic results in isentropic but isentropic does not mean reversible adiabatic. A reversible adiabatic and an irreversible adiabatic process can have a common point because a cycle is possible with single reservoir with $W < 0$ & $Q < 0$

The immediate question for this construction of cycle comprising of 2 adiabatic and isothermal process is that, some body may argue the above cycle may be for a thermodynamic device which takes work (during adiabatic process) gives net heat output (during isothermal process), then it is not violating the 2nd law. But argument is that as the cycle is reversible then the reverse process must be possible which will violate the 2nd law. Hence the above argument is justified.

